

Cloud Computing & AWS Examination

Duration: 2 Hours

Total Marks: 50

NAME : Mandavi kumari

BATCH : 04

Instructions:

- Answer all questions.
 - Read each question carefully before answering.
 - Write SQL commands or AWS terminology where applicable.
 - Marks are indicated beside each question.
 - No electronic devices or internet access allowed.
-

Section A – Multiple Choice Questions (15 × 1 = 15 Marks)

Choose the correct option.

1. Which of the following is the main purpose of virtualization?

- A. Increase hardware size
- B. Run multiple virtual machines on a single physical machine
- C. Reduce RAM capacity
- D. Create databases

Ans:- B

2. Which service model provides virtual servers on demand?

- A. SaaS

B. PaaS

C. IaaS

D. FaaS

Ans:- C

3. AWS stands for

A. Amazon Web Storage

B. Amazon Web Services

C. Advanced Web Services

D. Amazon World Server

Ans:- B

4. Which AWS service is mainly used to launch virtual machines?

A. S3

B. EC2

C. IAM

D. RDS

Ans:- B

5. Which storage service provides object storage?

A. EBS

B. EC2

C. S3

D. RDS

Ans:- C

6. Which AWS service manages relational databases?

- A. Lambda
- B. RDS
- C. CloudFront
- D. SNS

Ans:-B

7. Which component allows communication between a VPC and the Internet?

- A. NAT Gateway
- B. Route Table
- C. Internet Gateway
- D. Security Group

Ans:-C

8. Which AWS component decides where network traffic should go?

- A. Route Table
- B. IAM
- C. RDS
- D. EC2

Ans:-A

9. Which subnet should contain publicly accessible web servers?

- A. Private Subnet
- B. Public Subnet
- C. Database Subnet

D. Reserved Subnet

Ans:-B

10. Which of the following is stateful?

A. Network ACL

B. Security Group

C. Route Table

D. CIDR Block

Ans:-B

11. Which of the following is stateless?

A. Security Group

B. IAM

C. Network ACL

D. Internet Gateway

Ans:-C

12. What does CIDR represent?

A. Database Identifier

B. IP Address Range

C. Storage Volume

D. User Permission

Ans:-B

13. Which IAM entity is used to define permissions?

A. Bucket

- B. Policy
- C. Subnet
- D. Snapshot

Ans:-B

14. Which gateway allows private instances to access the Internet without allowing inbound connections?

- A. Internet Gateway
- B. NAT Gateway
- C. VPN Gateway
- D. Direct Connect

Ans:- B

15. Which of the following is NOT an advantage of cloud computing?

- A. Scalability
- B. High Availability
- C. Pay-as-you-go Pricing
- D. Unlimited Free Hardware

Ans:- D

Section B – Very Short Answer Questions (10 × 2 = 20 Marks)

Answer in 1–2 lines.

16. Define virtualization.

Ans:- Virtualization is the technology of creating virtual versions of physical resources such as servers, storage, or operating systems on a single physical machine.

17. Mention any two limitations of a traditional IT infrastructure.

Answer:

- High hardware and maintenance costs.
 - Limited scalability.
-

18. What is cloud computing?

Answer:

Cloud computing is the delivery of computing services such as servers, storage, databases, and software over the internet on a pay-as-you-go basis.

19. Differentiate between Public Cloud and Private Cloud.

Answer:-

Public Cloud	Private Cloud
Owned by cloud providers	Owned by a single organization
Shared by multiple users	Dedicated to one organization
Lower cost	Higher security and control

20. What is the purpose of an IAM Role?

Answer:

An IAM Role provides temporary permissions to AWS services or users without requiring long-term credentials.

21. What is the difference between an EC2 instance and an RDS instance?

EC2: A virtual server used to run applications.

RDS: A managed relational database service for storing and managing databases.

22. What is the purpose of a Security Group?

Answer:

A Security Group acts as a virtual firewall that controls inbound and outbound traffic for AWS resources.

23. What is a CIDR block?

Answer:

A CIDR block defines a range of IP addresses used within a VPC or subnet.

24. Why is a Route Table required inside a VPC?

Answer:

A Route Table determines how network traffic is routed between subnets, the internet, and other networks.

25. What is the purpose of an S3 Bucket?

Answer:

An S3 Bucket is used to store and retrieve objects such as files, images, videos, and backups in AWS.

Section C – Short Answer Questions (3 × 3 = 9 Marks)

Answer briefly.

26. Explain any three advantages of cloud computing.

(3 Marks)

Answer:

The Three advantage of cloud computing are:-

1. Scalability: Cloud resources can be increased or decreased based on demand.
 2. Cost-Effective: Users pay only for the resources they use, reducing hardware and maintenance costs.
 3. High Availability: Cloud services provide reliable access with minimal downtime through multiple data centers.
-

27. Differentiate between a Security Group and a Network ACL.

(3 Marks)

Ans:-

Security Group	Network ACL (NACL)
Stateful firewall	Stateless firewall
Applied at the instance level	Applied at the subnet level
Allows only Allow rules	Supports both Allow and Deny rules

28. Explain the architecture of a VPC containing:

- Public Subnet
- Private Subnet
- Internet Gateway
- NAT Gateway
- Route Tables

Draw a neat diagram if necessary.

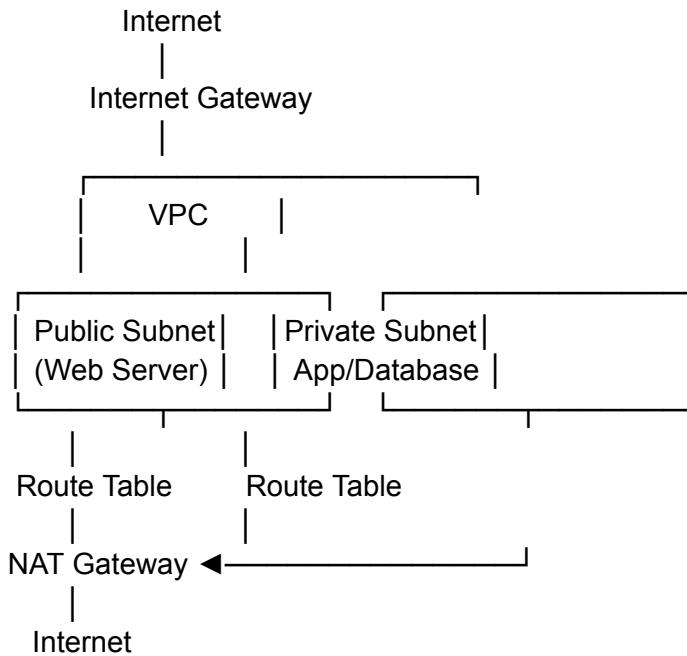
(3 Marks)

Answer:

A Virtual Private Cloud (VPC) is a private network in AWS that contains public and private subnets.

- Public Subnet: Hosts web servers that are accessible from the internet.
- Private Subnet: Hosts databases or application servers that are not directly accessible from the internet.
- Internet Gateway (IGW): Connects the public subnet to the internet.
- NAT Gateway: Allows instances in the private subnet to access the internet for updates without allowing inbound internet connections.

- Route Tables: Define how network traffic is routed between the subnets, Internet Gateway, and NAT Gateway.
- Diagram:-



Section D – Long Answer Question (1 × 6 = 6 Marks)

29.

A company wants to host an online shopping website on AWS.

Design a secure AWS architecture using the services studied in class.

Your answer should include:

- VPC
- Public and Private Subnets
- Internet Gateway
- NAT Gateway
- Route Tables
- Security Groups
- EC2
- RDS

- S3
- IAM

Explain the role of each service and describe how they interact to build a secure, highly available application.

(6 Marks)

Answer:

A secure AWS architecture for an online shopping website consists of the following components:

1. VPC (Virtual Private Cloud)

- A VPC is a private virtual network that hosts all AWS resources securely.
- It isolates the application from other AWS customers.

2. Public Subnet

- The Public Subnet contains the EC2 Web Server.
- It has a route to the Internet Gateway, allowing customers to access the shopping website.

3. Private Subnet

- The Private Subnet contains the RDS Database.
- It is not directly accessible from the internet, improving security.

4. Internet Gateway (IGW)

- The Internet Gateway connects the VPC to the internet.
- It allows users to access the EC2 web server in the public subnet.

5. NAT Gateway

- The NAT Gateway is placed in the public subnet.
- It allows resources in the private subnet (such as RDS or private EC2 instances) to access the internet for updates without allowing inbound internet traffic.

6. Route Tables

- Public Route Table: Routes internet traffic (0.0.0.0/0) to the Internet Gateway.
- Private Route Table: Routes internet-bound traffic through the NAT Gateway.

7. Security Groups

- Security Groups act as virtual firewalls.
- Web Server (EC2):
 - Allow HTTP (80)
 - Allow HTTPS (443)
 - Allow SSH (22) only from the administrator's IP.
- Database (RDS):
 - Allow database access only from the EC2 instance.

8. EC2 (Elastic Compute Cloud)

- Hosts the online shopping website.
- Receives customer requests and communicates with the database.

9. RDS (Relational Database Service)

- Stores customer information, orders, and product details.
- Located in the private subnet for enhanced security.

10. S3 (Simple Storage Service)

- Stores product images, invoices, backups, and static website files.
- Provides scalable and durable object storage.

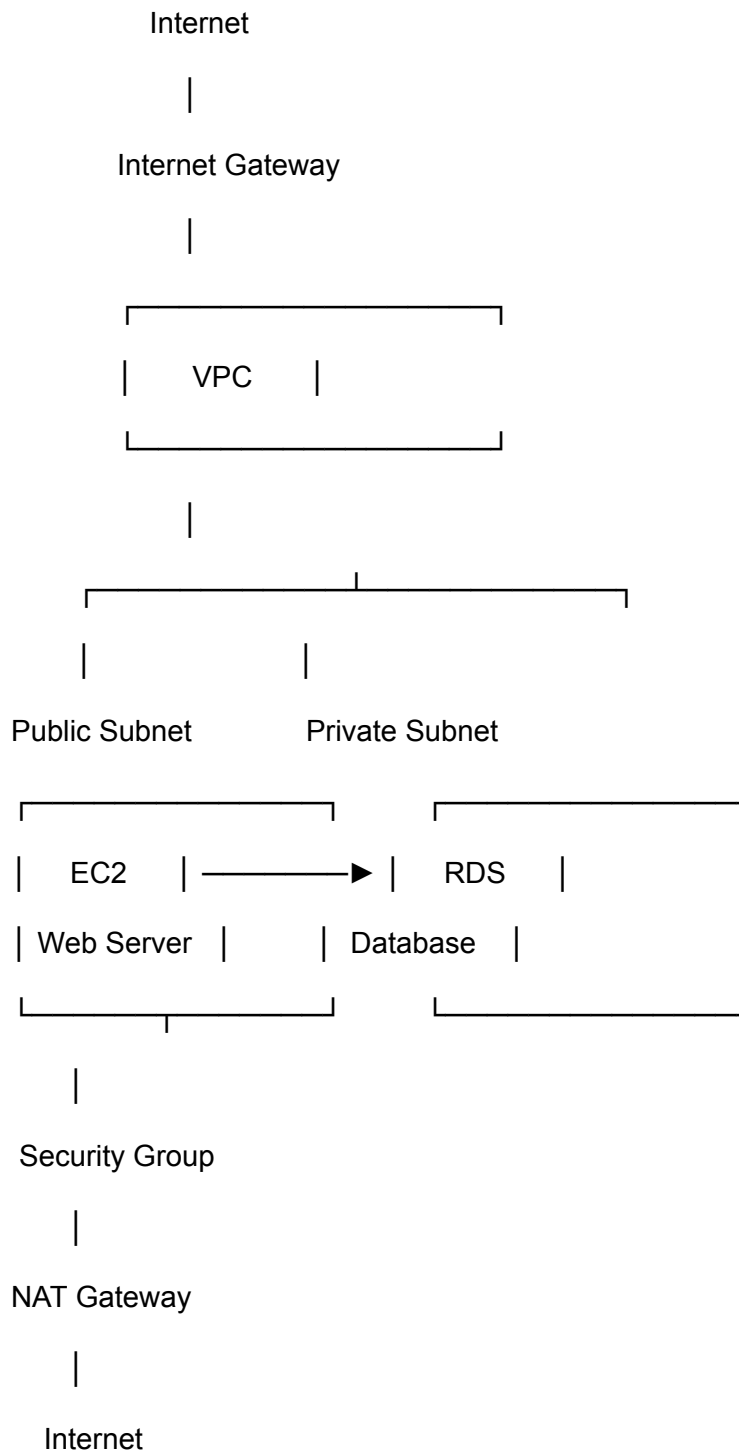
11. IAM (Identity and Access Management)

- Controls user and service permissions.
- Ensures only authorized users and AWS services can access resources.

Working of the Architecture:-

1. A customer opens the shopping website through the internet.
2. The request reaches the Internet Gateway and is forwarded to the EC2 Web Server in the Public Subnet.
3. The EC2 instance processes the request and retrieves or stores data in the RDS Database located in the Private Subnet.
4. Product images and static files are stored in S3.
5. Route Tables direct network traffic correctly within the VPC.
6. Security Groups protect EC2 and RDS by allowing only required traffic.
7. IAM manages secure access to AWS resources.
8. The NAT Gateway enables the private subnet to download updates without exposing it to incoming internet traffic.

12. Architecture Diagram:-



S3 Bucket (Images & Backups)

IAM (User & Resource Permissions)